
Rope Flow Music

An Embodied Instrument that Sonifies a Movement Practice

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Abstract

We present an unsupervised learning approach for turning *rope flow* (a movement practice in which a weighted rope is swung in continuous, looping patterns around the body) into a musical instrument. Two inertial sensors mounted on the handles of a rope stream the motion to a machine-learning pipeline that discovers the vocabulary of movements a performer produces, and maps that vocabulary to sound in real time. The artwork is a performance: a body swinging a rope, generating music whose rhythm comes from the swing rather than a clock. It addresses the theme of *Agency* by asking where agency lives when music is made together by a body, a learned model, and a language model coupled through an action-perception loop to produce an artistic work. We propose a *symbiotic* rather than extractive relationship with the machine, one in which the performer trains the model with their own movement and the machine reveals structure in that movement the performer cannot see.

1 The work

A person stands with a weighted rope and begins to swing it — underhand, overhand, the matador, the sneaks — in the continuous, rotational patterns of *rope flow*, a movement practice that grew out of David Weck’s rotational training and is now practiced worldwide. As they move, music emerges: chords change on the turn of each swing, and the timbre gets modulated by the energy, speed and force of the motion. The gravitational forces and moments that constrain the coupling of rope and flow artist create rhythm, naturally sonifiable. The rope swings at roughly one cycle per second, and that frequency *is* the tempo; speeding up, slowing down, and holding are expressive choices made with the whole body.

The submitted three-minute video shows the movement and the latent space side by side. Here the latent representation, which is itself a form of visual art, is the representation that the artificial agents has of the human agent. It is that representation that is being clustered and mapped to sound, back to the human who gets to evaluate the music, react to it physically, and propose adjustments (via natural language) to the machine. The human acts as both instrument designer and musician.

A python code both classifies movements, and maps them to midi messages that are sent in real-time to a DAW (here using Ableton Live). An average energy signal filters the wavetable sound used in this demo, and the total acceleration is used to control the volume, creating the spacious pulsating sound.

The instrument is deliberately used as a *constraint*. Free-form gesture-to-sound systems face an unbounded space of movements with no natural correspondence to music. Rope flow instead supplies a bounded, periodic, learnable vocabulary; a handful of named movements a performer can practice and improve at, exactly as one improves at an instrument. That constraint is what makes the mapping from movement to sound tractable, and what lets the piece be *performed* rather than merely

37 triggered, enabling a constructive collaboration between a machine equipped with a LLM that builds
38 the models and the interfaces, and a human who refines their movement and musical practice.

39 **2 The roles of AI and ML**

40 The main problem is to align two structured domains — a continuous, quasi-periodic, twelve-
41 channel motion signal (3D accelerometer and 3D gyroscope on each hand), and a stream of musical
42 events and expressive parameters; when their units, dimensions, and natural segmentation all differ
43 and no paired (movement, music) supervision exists. The system addresses this in three ways.

44 **Unsupervised vocabulary discovery.** We state a few approaches that have been used for latent
45 space discovery and clustering. A vector-quantized autoencoder learns, from the raw motion alone,
46 a small codebook of recurring movement primitives; a parallel representation models each cyclic
47 movement as a closed loop in a low-dimensional latent space, whose phase becomes a continuous
48 control signal. Fitting a sparse dynamical model to that loop recovers a Hopf oscillator whose
49 frequency matches the measured swing frequency. The instrument’s physics reappears, inside the
50 learned model. The model is unsupervised by necessity: the practice begins at the rope but is meant
51 to grow, and a fixed set of labels would freeze a vocabulary that performers are meant to extend.
52 This ensures that the collaboration is creative: enabling the human to discover new movements
53 rather than be constrained by designed ones.

54 **The machine sees what the body cannot.** Because the representation is learned from movement
55 rather than imposed, it surfaces structure the performer did not plan. In one session, a “near-steer /
56 far-steer” distinction — looking outward versus inward through the rope — emerged on its own as a
57 principal axis of the latent space. The performer then learns about their own movement by reading
58 it back from the machine: a two-way exchange, not a one-way command.

59 **Matched-complexity mapping.** Rather than wire one sensor to one knob, the mapping connects
60 movement and musical features of comparable rate, dimension, and count: a swing frequency near
61 1 Hz to musical tempo; a vocabulary of a dozen movement primitives to the order of a diatonic chord
62 set; bilateral coordination between the hands to harmonic motion. Naming is optional and cheap —
63 a lightweight classifier attaches semantic labels only where a performer wants them — so discovery
64 stays unsupervised while control stays legible.

65 **3 Agency**

66 The piece is built around a question the theme names directly: when the rope makes music, who is
67 acting?

68 **Time returned to the body.** Most digital instruments offload timekeeping to a clock, and quietly
69 remove time as an expressive dimension. Soniqflow gives it back: rhythm is generated by a swinging
70 mass with real physical feedback (a counterweight the performer works with and against) so tempo
71 becomes something the whole body shapes. The expressive stretching and contracting of time,
72 central to many musical traditions, is restored as a physical act.

73 **Many agents, not one.** What comes out of the instrument is shaped by the performer’s move-
74 ment, by the designer who chose the mappings and voices, by the learned model that discovered the
75 vocabulary.

76 **Symbiosis instead of extraction.** Much of our present relationship with machines is extractive:
77 systems built to capture attention, trained on work taken without consent. Soniqflow proposes the
78 opposite. The model is trained on the performer’s *own* movement, recorded as they play, rather than
79 on scraped data; the exchange is meant to make both sides better through low-quantity, high-quality
80 interaction rather than extraction. The performer trains the machine with their body, the machine
81 reveals the body to the performer, and the two co-evolve; a coupling closer to a duet than to a tool
82 used or a user mined. In this sense the work is also a small argument about ownership: what matters
83 is not possessing the output but keeping the relationship healthy and the practice open to others.

84 **4 Author biography**

85 Joseph Bakarji is an Assistant Professor in the Department of Mechanical Engineering, and the
86 Artificial Intelligence, Data Science and Computing Hub, at the American University of Beirut.
87 He received his PhD at Stanford University in 2020, working under Daniel Tartakovsky, where he
88 developed multiscale stochastic models for granular materials, receiving the Henry J. Ramey, Jr.
89 and the Frank G. Miller fellowship awards. Subsequently, he spent three years as a postdoctoral
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